

F1 in Schools Car Design Report

Roshan J
23BTT034

1. Introduction

This project focuses on the design and development of an F1 in Schools model race car using Autodesk Fusion 360. The primary objective of the project is to create a lightweight, aerodynamic, and high-performance miniature Formula 1 car suitable for competition and engineering analysis.

The design process involved CAD modeling, aerodynamic consideration, dimensional accuracy, and structural optimization. The completed design demonstrates the application of engineering principles, computer-aided design (CAD), and product development techniques.

2. Objectives

The main objectives of the project are:

- To design a high-speed F1 in Schools race car.
 - To create a streamlined aerodynamic body.
 - To reduce drag and improve stability.
 - To ensure proper wheel alignment and structural balance.
 - To develop a manufacturable CAD model using Fusion 360.
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3. Software Used

Autodesk Fusion 360

Fusion 360 was used for:

- 3D modeling
- Sketch creation
- Extrusion and surface modeling
- Assembly design

- Rendering and visualization
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4. Design Methodology

The following steps were followed during the design process:

4.1 Sketch Creation

Initial 2D sketches were created to define:

- Car body profile
- Wheel positions
- Aerodynamic shape
- Overall dimensions

4.2 3D Modeling

The 2D sketches were converted into a 3D model using:

- Extrude features
- Fillets and chamfers
- Surface modifications
- Symmetry tools

4.3 Aerodynamic Optimization

The body shape was designed with smooth curves and tapered surfaces to:

- Reduce air resistance
- Improve speed
- Enhance airflow around the car

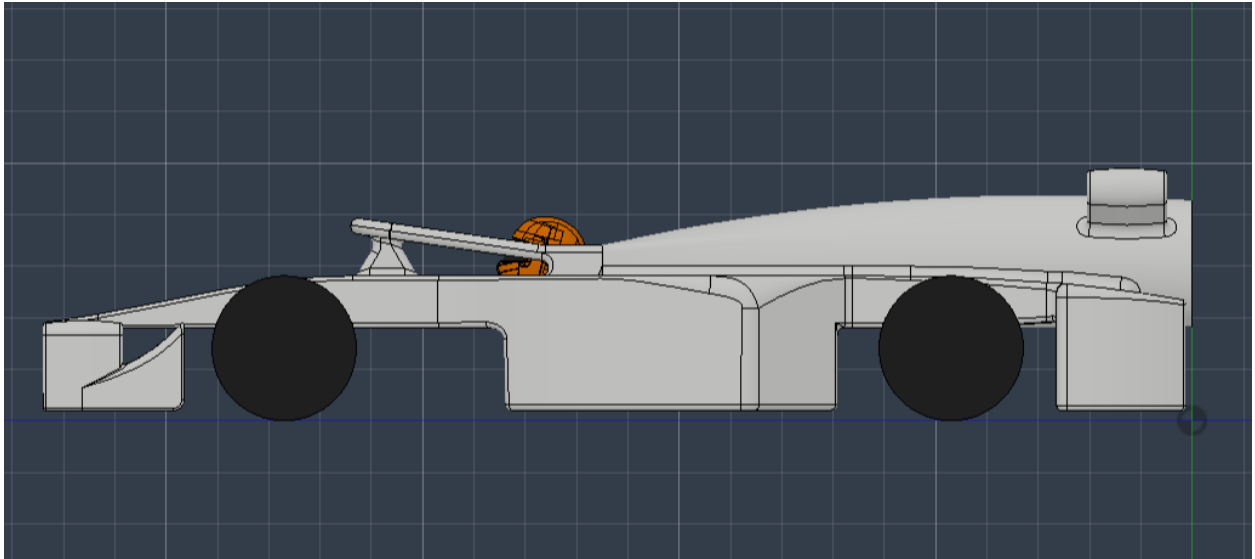
4.4 Wheel and Axle Placement

Proper wheel positioning was considered to:

- Maintain balance
 - Improve stability
 - Reduce rolling resistance
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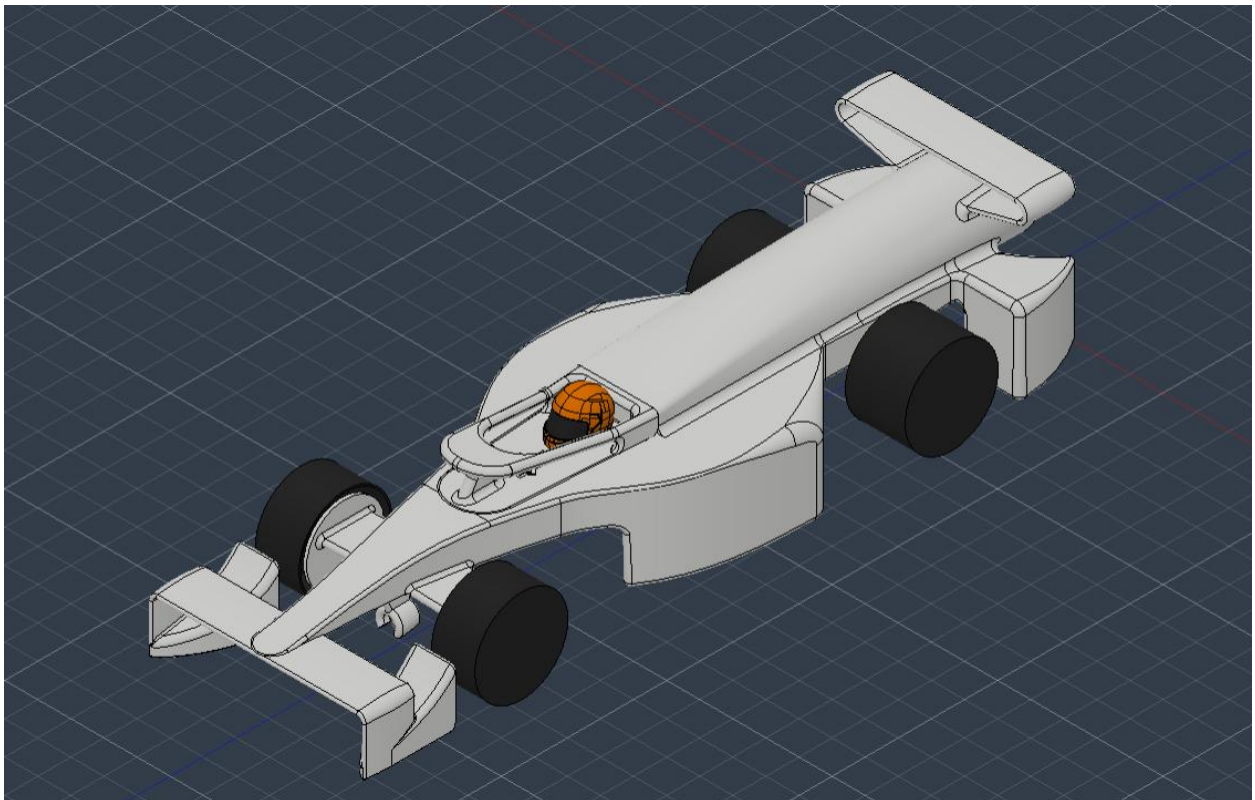
5. Design Images

Side View of F1 Car Design



Side View of F1 Car

Isometric View of F1 Car Design



Isometric View of F1 Car

5. Design Features

Key features of the designed F1 car include:

- Streamlined aerodynamic body
 - Lightweight structure
 - Smooth surface finishing
 - Optimized wheelbase
 - Compact and efficient design
 - Symmetrical body geometry
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6. Advantages of the Design

- Reduced aerodynamic drag
 - Improved speed performance
 - Better stability during racing
 - Efficient use of materials
 - Professional CAD-based design approach
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7. Applications

This project can be used for:

- F1 in Schools competitions
 - CAD and engineering education
 - Aerodynamic analysis studies
 - Prototype manufacturing
 - Product design learning
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8. Challenges Faced

During the design process, the following challenges were encountered:

- Maintaining aerodynamic efficiency
- Balancing strength and weight
- Ensuring dimensional accuracy
- Designing smooth body curves
- Achieving proper wheel alignment

These challenges were solved through iterative CAD modifications and optimization.

9. Future Improvements

Possible future improvements include:

- Advanced aerodynamic simulation
 - Weight reduction optimization
 - Material analysis
 - CFD (Computational Fluid Dynamics) testing
 - Improved wheel and axle systems
 - Enhanced body aesthetics
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10. Conclusion

The F1 in Schools car model was successfully designed using Autodesk Fusion 360. The project demonstrates the practical implementation of CAD modeling, aerodynamic principles, and engineering design techniques. The final model provides a strong foundation for racing performance, further analysis, and manufacturing.

The project enhanced skills in 3D modeling, design optimization, and engineering problem-solving.

11. References

1. Autodesk Fusion 360 Documentation
2. F1 in Schools Technical Regulations
3. CAD Design Tutorials and Engineering Resources
4. Aerodynamics and Vehicle Design Principles